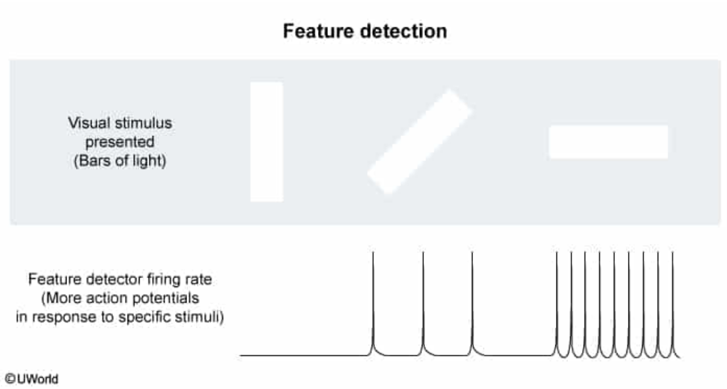
In an experiment, researchers find that certain neurons in the visual cortex preferentially fire in response to a bar of light that is oriented at a specific angle, and that different neurons similarly respond to bars of light oriented at different angles. This finding provides the strongest evidence for:

- ✔ ◉ A. feature detection. [61%]
- ○ B. parallel processing. [16%]
- ○ C. spreading activation. [12%]
- ○ D. sensory adaptation. [10%]

Correct
61% answered correctly

Explanation:



Feature detection involves the perceptual **discrimination** of **specific aspects** of a given **stimulus** via feature detectors. **Feature detectors** are **specific neurons** that **preferentially fire** in response to very **specific stimuli**.

Feature detection occurs for all the senses but is most often described regarding vision. Feature detectors in the visual system respond to aspects of the visual stimulus, such as horizontal lines or right angles (among others). The visual system is organized such that feature detectors synapse on neurons that respond to more complex stimuli (eg, faces) localized in certain areas of the brain (eg, fusiform face area).

An experiment in which researchers find that certain neurons in the visual cortex preferentially fire in response to specific stimuli (eg, a bar of light oriented at a specific angle) provides the strongest evidence for *feature detection*.

(Choice B) Parallel processing describes the brain's ability to simultaneously process various components (eg, color, motion) of a visual stimulus. This experiment did not demonstrate the simultaneous firing of different neurons detecting different components of a stimulus, so it does not provide strong evidence for parallel processing.

(Choice C) Spreading activation occurs when a node (ie, concept) within an individual's semantic network (a uniquely organized cognitive web of information) triggers the activation of other, related nodes, a process known as priming. Because the experiment did not test priming, its findings do not provide evidence for spreading activation.

(Choice D) Sensory adaptation is a diminished nervous system response over time to stimuli that remain constant, resulting in diminished stimulus perception (eg, a constant odor becomes less noticeable over time). This experiment did not present stimuli until the response diminished, so its findings do not provide evidence for sensory adaptation.

Educational objective:

Feature detection involves feature detector neurons that preferentially fire in response to specific stimuli (eg, a right angle causes feature detectors in the visual system to respond).

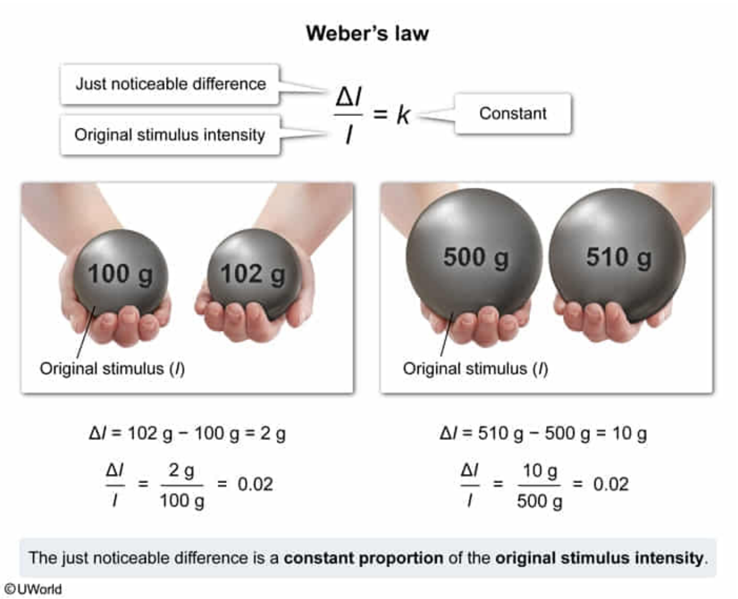
Time spent: 0 sec	**QID**:402323
Subject: Behavioral Sciences	**System:** Sensation, Perception, and Consciousness

A pharmaceutical company is testing how much sugar should be added to a pediatric oral liquid suspension to make the medication more palatable. Researchers determine that adding 0.2 g of sugar to 20 mL of oral suspension Z results in children detecting a just noticeable difference in taste. Based on Weber's law, should the addition of 0.2 g of sugar to 50 mL of oral suspension Z be detectable 50% of the time?

- A. Yes, because 0.2 g sugar is the just noticeable difference [12%]
- B. Yes, because Weber's law is based on a constant value [2%]
- C. No, because 0.2 g is below the absolute threshold [20%]
- ✔ D. No, because Weber's law is based on a ratio [63%]

Correct
63% answered correctly

Explanation:



The **smallest difference between two stimuli** a person can detect 50% of the time is called the **just noticeable difference (JND)**, or difference threshold. As the intensity of the original stimulus increases, the difference threshold also increases. For example, although the difference between weights of 100 and 102 g may be detectable, the difference between weights of 500 and 502 g is not, even though the difference is also 2 g.

This relationship is quantified by **Weber's law**, which states that the **ratio** of the **size of the JND** to the **original stimulus** intensity remains **constant**. In other words, the change in a stimulus needed to detect a difference is a constant proportion (ie, if one can just detect a difference between weights of 100 and 102 g, the difference between weights of 500 and 510 g would also be just detectable).

If the addition of 0.2 g of sugar to 20 mL of suspension Z is just noticeable 50% of the time, then for a volume that is 2.5 times bigger (20 mL × 2.5 = 50 mL), 0.5 g of sugar (0.2 g × 2.5 = 0.5 g) would be the JND. Therefore, the addition of 0.2 g of sugar to 50 mL of oral suspension Z would *not* be detectable because Weber's law is based on a ratio.

(Choice A) The addition of 0.2 g of sugar is the JND for 20 mL of suspension; for 50 mL of suspension the JND is 2.5 times greater, or 0.5 g. Therefore, anything less than 0.5 g should *not* be detectable 50% of the time.

(Choice B) Weber's law is based on a constant *proportion*, not a constant *value*, so the addition of 0.2 g of sugar to 50 mL of oral suspension Z should *not* be detectable 50% of the time.

(Choice C) The absolute threshold is the intensity value at which an individual can detect a stimulus 50% of the time; 0.2 g represents the *difference threshold*, or the smallest difference between two stimuli that can be detected 50% of the time, not the absolute threshold.

Educational objective:
Weber's law states that the proportion of the just noticeable difference (the smallest difference between two stimuli a person can detect 50% of the time) to the original stimulus intensity is a constant.